

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.9091199364437741, median 0.8088446441416963, std: 0.5327380797203257
Gyroscope error (imu0): mean 4.665363060084019e-10, median 1.2961838227966708e-11, std: 8.28101624860767e-09
Accelerometer error (imu0): mean 9.103320965170744e-10, median 2.644107024920545e-11, std: 1.624760870243052e-08

Residuals

Reprojection error (cam0) [px]: mean 0.9091199364437741, median 0.8088446441416963, std: 0.5327380797203257
Gyroscope error (imu0) [rad/s]: mean 1.1195180488959459e-12, median 3.110371402224127e-14, std: 1.9871437729757155e-11
Accelerometer error (imu0) [m/s^2]: mean 2.5748079943159597e-11, median 7.478664030017221e-13, std: 4.595517716621672e-10

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.00521702 -0.9999855 -0.00133811 0.00001705]
[-0.99998067 0.00521246 0.0033886 0.00005642]
[-0.00338157 0.00135577 -0.99999336 -0.00001808]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.00521702 -0.99998067 -0.00338157 0.00005645]
[-0.9999855 0.00521246 0.00135577 0.00001678]
[-0.00133811 0.0033886 -0.99999336 -0.00001824]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
0.018778119312694658

Gravity vector in target coords: [m/s^2]

[0.10320338 0.13922387 -9.80501855]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [478.70711455675183, 478.0180023752]

Principal point: [325.1324954902568, 230.9347645850685]

Distortion model: radtan

Distortion coefficients: [0.0413202957056939, -0.0350456928606942, -0.0011150828567765595, 0.0022116060179409626]

Type: aprilgrid

Tags:

Rows: 5

Cols: 4

Size: 0.044 [m]

Spacing 0.0044 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200.0

Accelerometer:

Noise density: 0.002

Noise density (discrete): 0.0282842712474619

Random walk: 0.003

Gyroscope:

Noise density: 0.00016968

Noise density (discrete): 0.0023996375726346676

Random walk: 1.9393e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

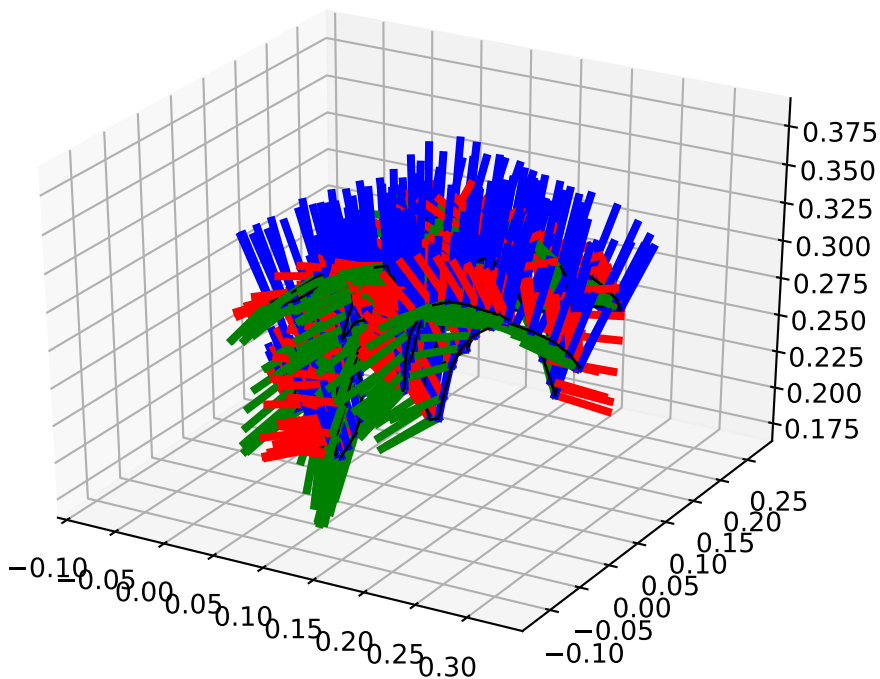
[0. 1. 0. 0.]

[0. 0. 1. 0.]

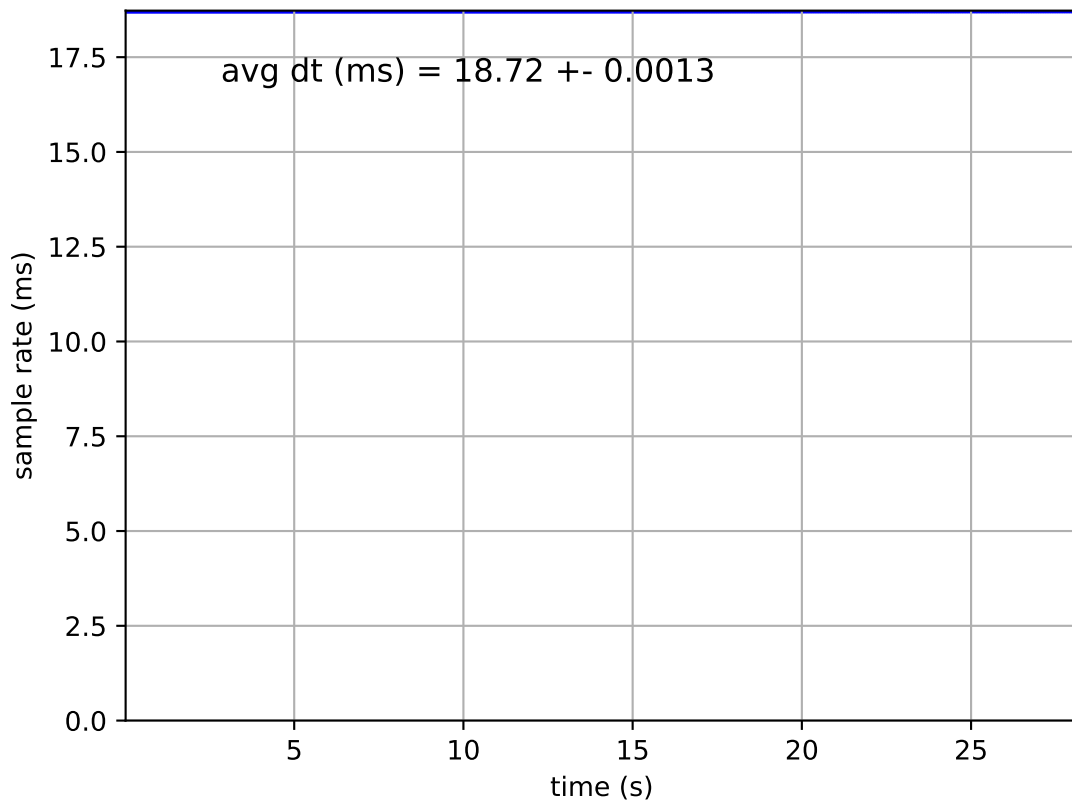
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

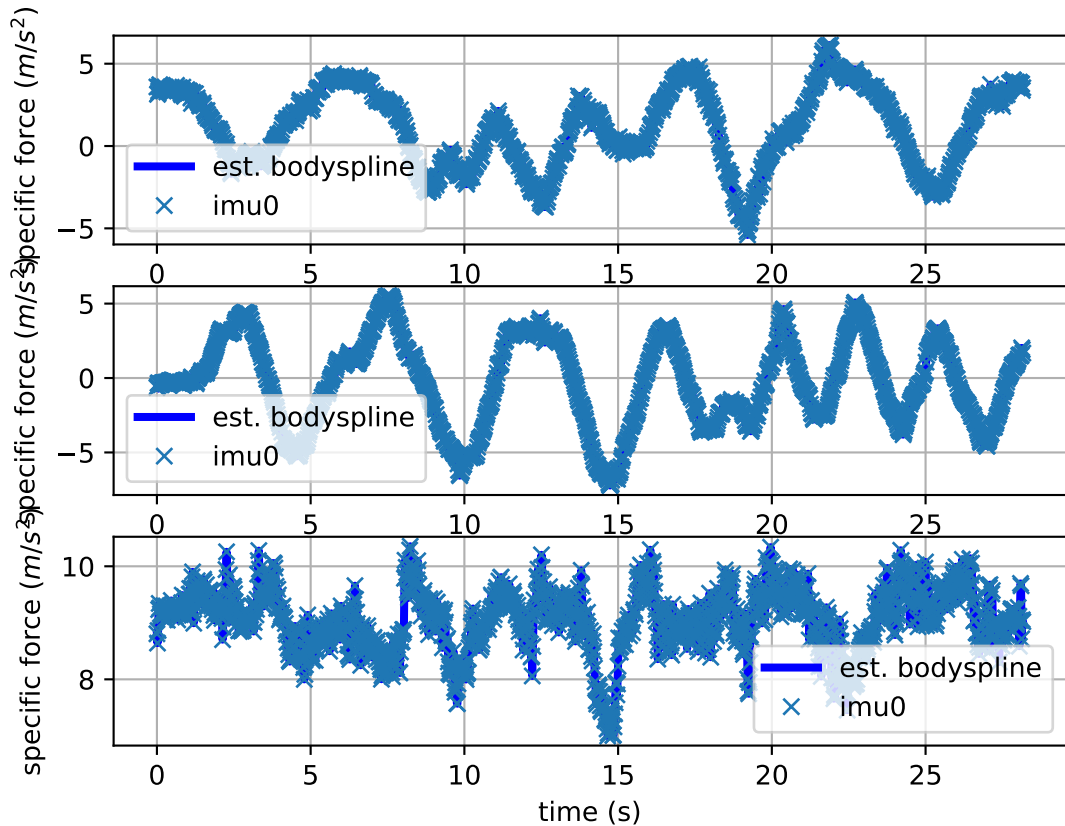
imu0: estimated poses



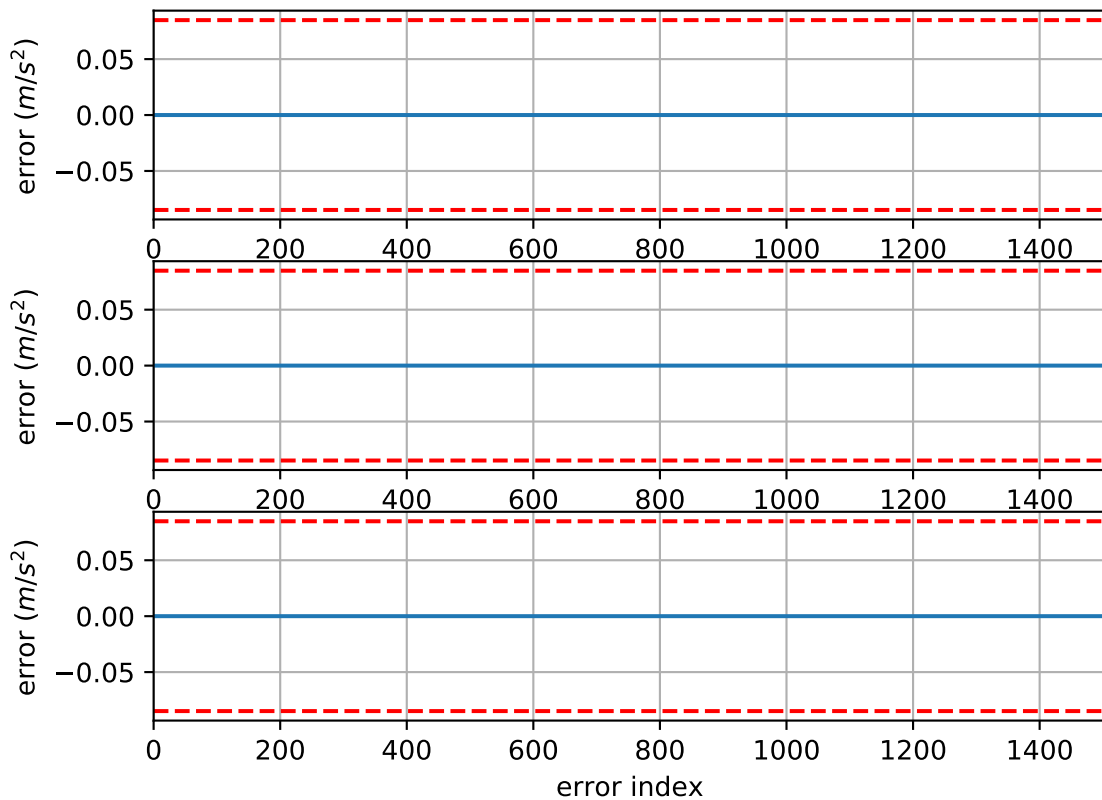
imu0: sample inertial rate



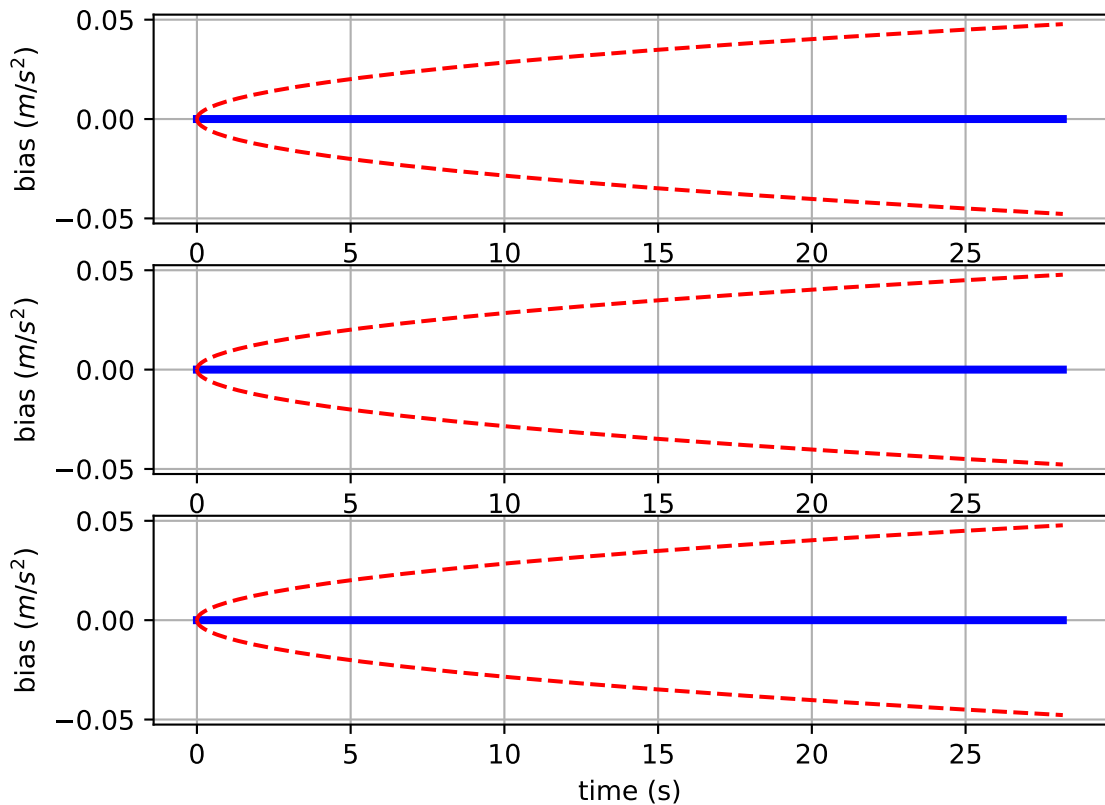
Comparison of predicted and measured specific force (imu0 frame)



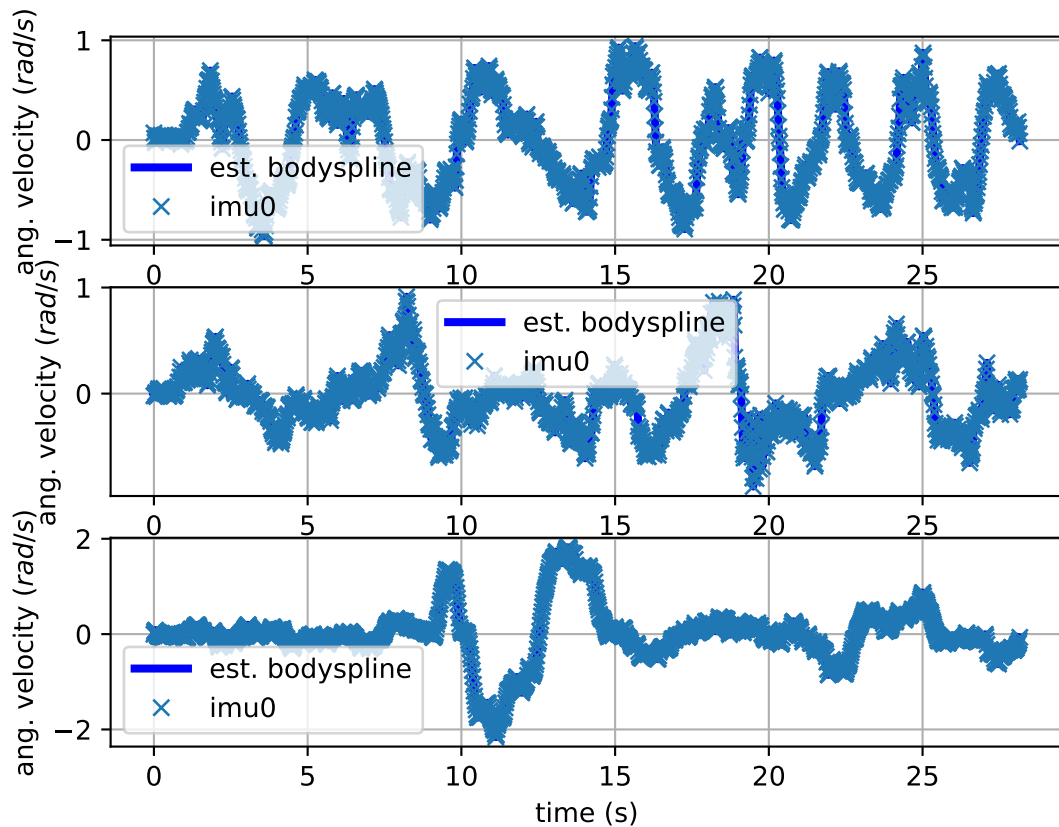
imu0: acceleration error



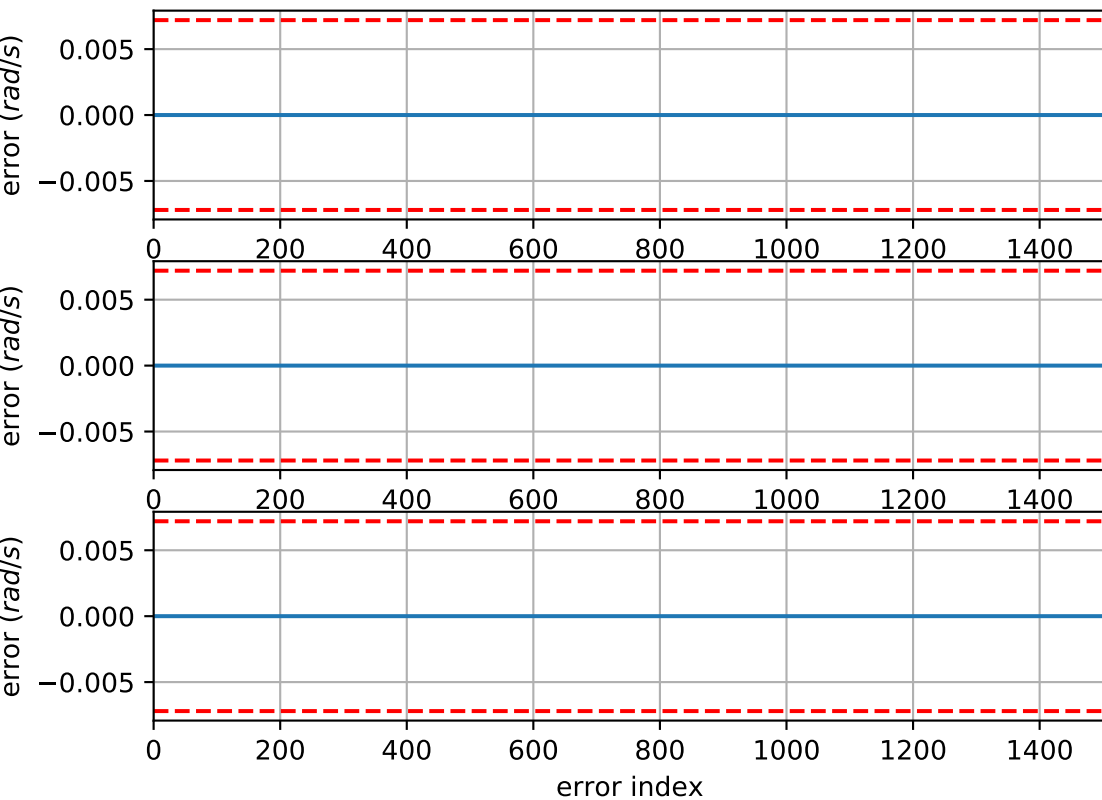
imu0: estimated accelerometer bias (imu frame)



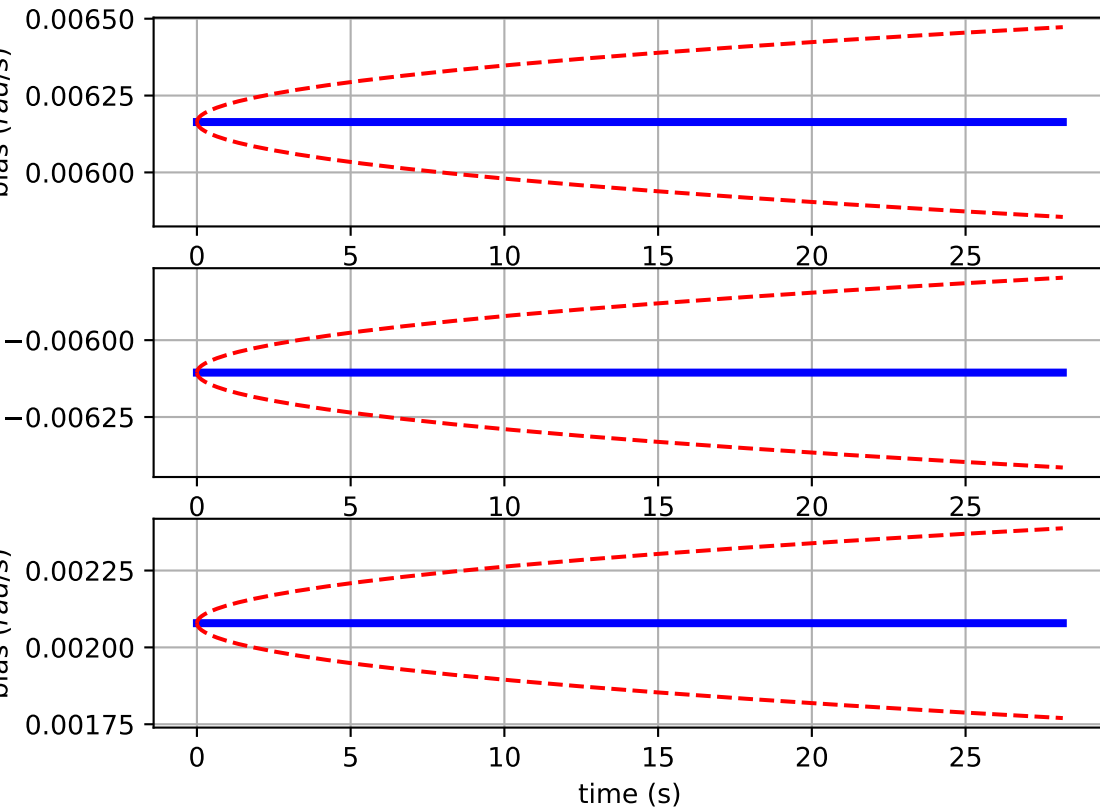
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

